

## 9.1 Decomposing the total sum of squares

Every sum of squares in a linear model is the squared length of a projection of the data. [Chapter 6](#) showed this for the residual sum of squares,  $\|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2$ , and for the drop in residual sum of squares between nested models,  $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2$  (Eq. 6.5.2). This section turns these observations into a general framework. An analysis of variance is a choice of mutually orthogonal subspaces that together fill the model space. Once that choice is made, the whole table follows from Pythagoras: the sums of squares add up because the pieces are orthogonal, and the degrees of freedom add up because dimensions of orthogonal subspaces add.

Throughout the chapter the model is  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$  with  $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$  (Definition 5.3.1).  $\mathbf{X}$  is  $n \times p$  of rank  $r$ , which may be less than  $p$ , and  $\mathbf{M}$  is the orthogonal projection onto  $\mathcal{C}(\mathbf{X})$ . The identities of the chapter hold for every data vector  $\mathbf{y}$ . Probability enters only where a covariance, an expectation or a distribution is stated, and systematically in [Section 9.6](#).

### 9.1.1 Orthogonal decompositions of the model space

#### Definition 9.1.1 — Analysis of variance decomposition

An **analysis of variance decomposition** of the model space is a list of subspaces  $\mathcal{V}_1, \dots, \mathcal{V}_k$  of  $\mathcal{C}(\mathbf{X})$  that are mutually orthogonal and whose sum is  $\mathcal{C}(\mathbf{X})$ :

$$\mathcal{C}(\mathbf{X}) = \mathcal{V}_1 \oplus \mathcal{V}_2 \oplus \dots \oplus \mathcal{V}_k, \quad \mathcal{V}_i \perp \mathcal{V}_j \ (i \neq j).$$

The subspace  $\mathcal{V}_{k+1} = \mathcal{C}(\mathbf{X})^\perp$  is the **error space**. Writing  $\mathbf{P}_i$  for the orthogonal projection onto  $\mathcal{V}_i$ , the **sum of squares** for  $\mathcal{V}_i$  is  $\|\mathbf{P}_i\mathbf{y}\|^2 = \mathbf{y}^\top \mathbf{P}_i \mathbf{y}$ , its **degrees of freedom** are  $r_i = \dim \mathcal{V}_i$ , and its **mean square** is  $\|\mathbf{P}_i\mathbf{y}\|^2 / r_i$ .

The same space can be decomposed in many ways, and the choice is the statistician's: the constants, a regressor of interest, a block of nuisance regressors, the linear and quadratic parts of a trend. The theorem collects what is true of all decompositions.

#### Theorem 9.1.1 — The analysis of variance identity

Let  $\mathcal{V}_1, \dots, \mathcal{V}_k$  be an analysis of variance decomposition of  $\mathcal{C}(\mathbf{X})$ , with projections  $\mathbf{P}_1, \dots, \mathbf{P}_k$  and dimensions  $r_1, \dots, r_k$ . Then:

- (a)  $\mathbf{P}_i \mathbf{P}_j = \mathbf{0}$  for  $i \neq j$ ,  $\mathbf{P}_1 + \dots + \mathbf{P}_k = \mathbf{M}$ , and

$$\mathbf{I} = \mathbf{P}_1 + \mathbf{P}_2 + \dots + \mathbf{P}_k + (\mathbf{I} - \mathbf{M}),$$

a sum of  $k + 1$  mutually orthogonal projections;

- (b) for every  $\mathbf{y} \in \mathbb{R}^n$  the vectors  $\mathbf{P}_1\mathbf{y}, \dots, \mathbf{P}_k\mathbf{y}, (\mathbf{I} - \mathbf{M})\mathbf{y}$  are mutually orthogonal and

$$\|\mathbf{y}\|^2 = \|\mathbf{P}_1\mathbf{y}\|^2 + \dots + \|\mathbf{P}_k\mathbf{y}\|^2 + \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2; \quad (9.1.1)$$

- (c)  $r_i = \text{rank}(\mathbf{P}_i) = \text{tr}(\mathbf{P}_i)$ ,  $r_1 + \dots + r_k = r$ , and the error space has dimension  $\text{tr}(\mathbf{I} - \mathbf{M}) = n - r$ ;

- (d) if the columns of  $\mathbf{Q}_i$  ( $n \times r_i$ ) form an orthonormal basis of  $\mathcal{V}_i$ , then  $\|\mathbf{P}_i \mathbf{y}\|^2 = \|\mathbf{Q}_i^\top \mathbf{y}\|^2$  is a sum of  $r_i$  squares, and together with an orthonormal basis  $\mathbf{Q}_{k+1}$  of the error space the matrix  $\mathbf{U} = [\mathbf{Q}_1, \dots, \mathbf{Q}_{k+1}]$  is orthogonal;
- (e) the subspaces  $\mathcal{W}_j = \mathcal{V}_1 \oplus \dots \oplus \mathcal{V}_j$  form a chain  $\mathcal{W}_1 \subseteq \dots \subseteq \mathcal{W}_k = \mathcal{C}(\mathbf{X})$  with projections  $\mathbf{N}_j = \mathbf{P}_1 + \dots + \mathbf{P}_j$ , and  $\mathbf{P}_j = \mathbf{N}_j - \mathbf{N}_{j-1}$  (with  $\mathbf{N}_0 = \mathbf{0}$ ). Conversely, every chain  $\{\mathbf{0}\} = \mathcal{W}_0 \subseteq \mathcal{W}_1 \subseteq \dots \subseteq \mathcal{W}_k = \mathcal{C}(\mathbf{X})$  gives an analysis of variance decomposition with  $\mathcal{V}_j = \mathcal{W}_j^\perp \cap \mathcal{W}_{j-1}$ .

*Proof.*

- (a) For any  $\mathbf{u}$ ,  $\mathbf{P}_j \mathbf{u} \in \mathcal{V}_j \subseteq \mathcal{V}_i^\perp$ , so  $\mathbf{P}_i \mathbf{P}_j \mathbf{u} = \mathbf{0}$  by Proposition 6.3.4(c). Applying Theorem 6.3.7 repeatedly, the sum  $\mathbf{P}_1 + \dots + \mathbf{P}_k$  is the orthogonal projection onto  $\mathcal{V}_1 \oplus \dots \oplus \mathcal{V}_k = \mathcal{C}(\mathbf{X})$ , and a projection is determined by its range, so the sum is  $\mathbf{M}$ . Each  $\mathbf{P}_i$  satisfies  $\mathbf{M} \mathbf{P}_i = \mathbf{P}_i$ , because its columns lie in  $\mathcal{C}(\mathbf{X})$ , and therefore  $(\mathbf{I} - \mathbf{M}) \mathbf{P}_i = \mathbf{0}$ . The projection  $\mathbf{I} - \mathbf{M}$  onto  $\mathcal{C}(\mathbf{X})^\perp$  (Theorem 6.3.3) is thus orthogonal to every  $\mathbf{P}_i$ .
- (b) Multiply  $\mathbf{I} = \sum_i \mathbf{P}_i + (\mathbf{I} - \mathbf{M})$  by  $\mathbf{y}$ . For two different pieces,  $(\mathbf{P}_i \mathbf{y})^\top (\mathbf{P}_j \mathbf{y}) = \mathbf{y}^\top \mathbf{P}_i \mathbf{P}_j \mathbf{y} = 0$ , using symmetry and (a). Pythagoras gives the identity.
- (c) Rank equals trace for projections (Proposition 6.3.4(b)), and the trace is additive, so  $\sum_i r_i = \text{tr}(\mathbf{M}) = r$  by Theorem 6.3.6, and  $\text{tr}(\mathbf{I} - \mathbf{M}) = n - r$ .
- (d) By Proposition 6.3.2,  $\mathbf{P}_i = \mathbf{Q}_i \mathbf{Q}_i^\top$ , so  $\mathbf{y}^\top \mathbf{P}_i \mathbf{y} = \|\mathbf{Q}_i^\top \mathbf{y}\|^2$ . Columns taken from different  $\mathbf{Q}_i$  are orthogonal because the subspaces are, so  $\mathbf{U}$  has orthonormal columns. It has  $r_1 + \dots + r_k + (n - r) = n$  of them, so it is square and hence orthogonal.
- (e) The  $\mathbf{N}_j$  are projections onto  $\mathcal{W}_j$  by Theorem 6.3.7, and the chain is nested by construction. Conversely, given a chain, Theorem 6.5.1(b) shows that  $\mathbf{N}_j - \mathbf{N}_{j-1}$  is the projection onto  $\mathcal{V}_j = \mathcal{W}_j^\perp \cap \mathcal{W}_{j-1}$ . For  $i < j$  we have  $\mathcal{V}_i \subseteq \mathcal{W}_i \subseteq \mathcal{W}_{j-1}$ , which is orthogonal to  $\mathcal{V}_j$ . The projections telescope to  $\mathbf{N}_k = \mathbf{M}$ , so by Theorem 6.3.7 the orthogonal sum of the  $\mathcal{V}_j$  is  $\mathcal{C}(\mathbf{M}) = \mathcal{C}(\mathbf{X})$ .

■

Part (e) says that decompositions and chains are the same thing. Listing the rows of an analysis of variance table from the top is the same as fitting a sequence of ever larger models and recording what each step adds. [Section 9.3](#) builds on this.

## 9.1.2 Degrees of freedom are dimensions

“Degrees of freedom” can seem like a rule of thumb (“subtract one for every parameter estimated”). Part (d) of [Theorem 9.1.1](#) gives the phrase an exact meaning. Rotate the data into the orthonormal coordinates  $\mathbf{z} = \mathbf{U}^\top \mathbf{y}$ . The rotation preserves length, and it splits  $\mathbf{z}$  into consecutive blocks  $\mathbf{Q}_1^\top \mathbf{y}, \dots, \mathbf{Q}_{k+1}^\top \mathbf{y}$  of sizes  $r_1, \dots, r_k, n - r$ . The sum of squares for  $\mathcal{V}_i$  is the sum of the squares of the  $r_i$  coordinates in its block. *A sum of squares has  $r_i$  degrees of freedom because it is a sum of  $r_i$  squares of orthonormal linear combinations of the data, and it cannot be written with fewer.*

The rotation also explains why degrees of freedom govern the size of a sum of squares. If  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , the coordinates  $\mathbf{U}^\top \mathbf{Y}$  are uncorrelated with variance  $\sigma^2$  (Theorem 2.2.1), so a block of  $r_i$  of them has expected sum of squares  $r_i \sigma^2$  plus the squared length of its mean

(Section 9.6). The divisor  $n - 1$  of the sample variance is the case  $\mathbb{R}^n = \text{span}(\mathbf{1}) \oplus \mathbf{1}^\perp$ : the  $n$  deviations  $y_i - \bar{y}$  lie in an  $(n - 1)$ -dimensional subspace, and the Helmert matrix of Example (Helmert matrices) writes their sum of squares as  $n - 1$  squared orthonormal contrasts.

### 9.1.3 The regression analysis of variance table

The most common decomposition comes from the chain  $\text{span}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{X})$ , available whenever  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ , which is the case for every model with an intercept. Write  $\mathbf{P}_1 = n^{-1}\mathbf{1}\mathbf{1}^\top$  for the projection onto the constant vectors (Example (Projections onto a line)). The chain gives the three-way decomposition of Eq. 6.5.1,

$$\mathbf{I} = \mathbf{P}_1 + (\mathbf{M} - \mathbf{P}_1) + (\mathbf{I} - \mathbf{M}),$$

with ranks 1,  $r - 1$  and  $n - r$ . The squared lengths are

$$\|\mathbf{P}_1\mathbf{y}\|^2 = n\bar{y}^2, \quad \text{SSR} = \|(\mathbf{M} - \mathbf{P}_1)\mathbf{y}\|^2 = \sum_i (\hat{y}_i - \bar{y})^2, \quad \text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \sum_i (y_i - \hat{y}_i)^2.$$

They are usually arranged as follows.

| Source                      | df      | Sum of squares               | Mean square                |
|-----------------------------|---------|------------------------------|----------------------------|
| Mean                        | 1       | $n\bar{y}^2$                 | $n\bar{y}^2$               |
| Regression (after the mean) | $r - 1$ | SSR                          | $\text{SSR}/(r - 1)$       |
| Residual                    | $n - r$ | SSE                          | $s^2 = \text{SSE}/(n - r)$ |
| Total                       | $n$     | $\mathbf{y}^\top \mathbf{y}$ |                            |

This is the **uncorrected** table. Its total is the raw sum of squares  $\sum_i y_i^2$ , with  $n$  degrees of freedom. The row for the mean is rarely of interest, because it records only that the average response is not zero. Deleting it gives the **corrected** table, whose total is  $\text{SST} = \sum_i (y_i - \bar{y})^2 = \|(\mathbf{I} - \mathbf{P}_1)\mathbf{y}\|^2$  on  $n - 1$  degrees of freedom. The corrected identity  $\text{SST} = \text{SSR} + \text{SSE}$  is Eq. 6.7.1. “Correcting for the mean” is thus a projection. We first project the data onto  $\mathbf{1}^\perp$  and then decompose what is left.

The ratio of the two mean squares,

$$F = \frac{\text{SSR}/(r - 1)}{\text{SSE}/(n - r)},$$

compares the variation along the regression directions with the variation in the error space. Under normality and when no regressor matters, it has the  $F(r - 1, n - r)$  distribution (Theorem 4.4.6). Section 9.6 explains why this ratio is the right comparison, and Chapter 11 develops the test.

A few formulas make the table easy to compute from any least squares solution.

#### Proposition 9.1.2 — Computing the sums of squares

Let  $\hat{\boldsymbol{\beta}}$  be any least squares estimate. Then:

- (a)  $\|\mathbf{M}\mathbf{y}\|^2 = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$  and  $\text{SSE} = \mathbf{y}^\top \mathbf{y} - \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$ ;
- (b) if  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ , then  $\text{SSR} = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y} - n\bar{y}^2$ ;

- (c) if  $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$  has full column rank and  $\tilde{\mathbf{X}}_1 = (\mathbf{I} - \mathbf{P}_1)\mathbf{X}_1$  is the matrix of centred regressors, then  $\text{SSR} = \hat{\boldsymbol{\beta}}_1^\top \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1 \hat{\boldsymbol{\beta}}_1 = \hat{\boldsymbol{\beta}}_1^\top \tilde{\mathbf{X}}_1^\top \mathbf{y}$ , where  $\hat{\boldsymbol{\beta}}_1$  holds the slopes.

*Proof.*

- (a)  $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{M}\mathbf{y}$  (Theorem 6.4.1), so  $\|\mathbf{M}\mathbf{y}\|^2 = \mathbf{y}^\top \mathbf{M}\mathbf{y} = \mathbf{y}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$ . The second formula follows from  $\text{SSE} = \mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{M}\mathbf{y}$  (Proposition 6.4.3). (b)  $\mathbf{M} - \mathbf{P}_1$  is a projection (Theorem 6.5.1), so  $\text{SSR} = \mathbf{y}^\top \mathbf{M}\mathbf{y} - \mathbf{y}^\top \mathbf{P}_1 \mathbf{y} = \|\mathbf{M}\mathbf{y}\|^2 - n\bar{y}^2$ ; now use (a). (c) By Lemma 6.6.1 with  $\mathbf{X}_2 = \mathbf{1}$ ,  $\mathbf{M} - \mathbf{P}_1$  is the projection onto  $\mathcal{C}(\tilde{\mathbf{X}}_1)$ , and by Theorem 6.6.2 the slopes are the coefficients of  $(\mathbf{M} - \mathbf{P}_1)\mathbf{y}$  in terms of the columns of  $\tilde{\mathbf{X}}_1$ . So  $(\mathbf{M} - \mathbf{P}_1)\mathbf{y} = \tilde{\mathbf{X}}_1 \hat{\boldsymbol{\beta}}_1$  and  $\text{SSR} = \|\tilde{\mathbf{X}}_1 \hat{\boldsymbol{\beta}}_1\|^2$ . The last equality uses  $\tilde{\mathbf{X}}_1^\top \mathbf{y} = \tilde{\mathbf{X}}_1^\top (\mathbf{M} - \mathbf{P}_1)\mathbf{y} = \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1 \hat{\boldsymbol{\beta}}_1$ . ■

In (a),  $\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$  is the same for every solution of the normal equations, even though  $\hat{\boldsymbol{\beta}}$  is not unique when  $r < p$ . That is no accident. It equals  $\|\mathbf{M}\mathbf{y}\|^2$ , which depends only on the model space.

#### Example. (An analysis of variance table for murder rates)

Return to the regression of the 2009 murder rate of the 50 US states on poverty, single parenthood and urbanization, with an intercept (Example (Poverty and murder rates)). Here  $r = 4$  and  $\bar{y} = 4.514$ . The listing computes every sum of squares from the coordinates of  $\mathbf{y}$  in an orthonormal basis of  $\mathcal{C}(\mathbf{X})$ , as in Theorem 9.1.1(d): the first coordinate gives the mean, the next three the regression.

| Source              | df | Sum of squares | Mean square |
|---------------------|----|----------------|-------------|
| Mean                | 1  | 1018.81        | 1018.81     |
| Regression          | 3  | 182.80         | 60.93       |
| Residual            | 46 | 101.96         | 2.217       |
| Total (uncorrected) | 50 | 1303.57        |             |
| Total (corrected)   | 49 | 284.76         |             |

The regression mean square is 27.49 times the residual mean square. If none of the three regressors were related to the murder rate and the errors were normal, a ratio this large would occur with probability about  $2.5 \times 10^{-10}$ . The proportion of the corrected total that the regression accounts for is  $182.80/284.76 = 0.6419$ , the  $R^2$  of Theorem 6.7.2.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")

y = data["murder"].to_numpy()
n = len(y)
X = np.column_stack([np.ones(n), data["poverty"], data["single"], data["urban"]])
r = np.linalg.matrix_rank(X)

Q, _ = np.linalg.qr(X)                                # orthonormal basis of C(X); first column
```

```

    spans 1
z = Q.T @ y                                # coordinates of My in that basis
ss_mean = z[0] ** 2                         # n * ybar^2
ss_reg = np.sum(z[1:] ** 2)                 # ||(M - P1) y||^2, r - 1 squared coordinates
ss_res = y @ y - np.sum(z ** 2)             # ||(I - M) y||^2

rows = [("mean", 1, ss_mean), ("regression", r - 1, ss_reg),
        ("residual", n - r, ss_res), ("total", n, y @ y)]
for name, df, ss in rows:
    print(f"{name:11s} df = {df:3d}    SS = {ss:9.2f}    MS = {ss / df:8.3f}")
F = (ss_reg / (r - 1)) / (ss_res / (n - r))
print(f"corrected total = {ss_reg + ss_res:.2f},    F = {F:.2f}")

```

[Run this code online](#)

The QR factorization supplies the orthonormal basis of [Theorem 9.1.1\(d\)](#); its first column is  $\pm \mathbf{1}/\sqrt{n}$  because Gram–Schmidt starts with the intercept (Proposition 6.2.2). No  $n \times n$  projection matrix is formed.

## 9.1.4 Models without an intercept

The corrected table depends on  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ . Without it, the chain  $\text{span}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{X})$  is not available, and the corrected “identity” fails.

### Proposition 9.1.3

For any model matrix  $\mathbf{X}$ , with  $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$  and  $\hat{\mathbf{e}} = (\mathbf{I} - \mathbf{M})\mathbf{y}$ ,

$$\sum_i (y_i - \bar{y})^2 = \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i \hat{e}_i^2 - 2\bar{y} \sum_i \hat{e}_i.$$

The last term vanishes for every  $\mathbf{y}$  iff  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ .

*Proof.* Write  $\mathbf{y} - \bar{y}\mathbf{1} = (\hat{\mathbf{Y}} - \bar{y}\mathbf{1}) + \hat{\mathbf{e}}$  and expand the squared length. The cross term is  $2(\hat{\mathbf{Y}} - \bar{y}\mathbf{1})^\top \hat{\mathbf{e}} = 2\hat{\mathbf{Y}}^\top \hat{\mathbf{e}} - 2\bar{y}\mathbf{1}^\top \hat{\mathbf{e}} = -2\bar{y}\mathbf{1}^\top \hat{\mathbf{e}}$ , since  $\hat{\mathbf{Y}} \perp \hat{\mathbf{e}}$ . If  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ , then  $\mathbf{1}^\top \hat{\mathbf{e}} = 0$  (Proposition 6.4.3(b)). If  $\mathbf{1} \notin \mathcal{C}(\mathbf{X})$ , take  $\mathbf{y} = \mathbf{1}$ . Then  $\bar{y} = 1$  and  $\mathbf{1}^\top \hat{\mathbf{e}} = \mathbf{1}^\top (\mathbf{I} - \mathbf{M})\mathbf{1} = \|(\mathbf{I} - \mathbf{M})\mathbf{1}\|^2 > 0$ , so the last term is not zero. ■

Without an intercept the valid decomposition is the uncorrected one,  $\|\mathbf{y}\|^2 = \|\mathbf{M}\mathbf{y}\|^2 + \text{SSE}$ , whose regression row has  $r$  degrees of freedom. That is appropriate only when the mean response is known to vanish when every regressor is zero.

### Example. (Forcing the murder regression through the origin)

Refit the model of [Example \(An analysis of variance table for murder rates\)](#) without the intercept. The residual sum of squares rises from 101.96 to 177.73, so the fit is much worse. Yet the residuals now sum to  $-8.332$ , and the would-be corrected identity is off by  $-2\bar{y} \sum_i \hat{e}_i = 75.22$ : the corrected total 284.76 is not  $31.81 + 177.73 = 209.54$ .

The uncentred ratio  $\|\hat{\mathbf{Y}}\|^2/\|\mathbf{y}\|^2$ , which some software reports as  $R^2$  for such models, is 0.8637, much larger than the 0.6419 of the better model with an intercept (Exercise (B1)).

```
X0 = X[:, 1:] # drop the intercept column
b0, *_ = np.linalg.lstsq(X0, y, rcond=None)
e0 = y - X0 @ b0
ybar = y.mean()
sst = np.sum((y - ybar) ** 2)
ssr_star = np.sum((X0 @ b0 - ybar) ** 2)
sse0 = e0 @ e0
print(f"SST = {sst:.2f} but SSR* + SSE = {ssr_star + sse0:.2f}")
print(f"sum of residuals = {e0.sum():.3f}, -2 ybar * sum = {-2 * ybar * e0.sum():.2f}")
```

[Run this code online](#)

## 9.1.5 Coarsening and refining

**Coarsening** merges rows: replacing  $\mathcal{V}_1$  and  $\mathcal{V}_2$  by  $\mathcal{V}_1 \oplus \mathcal{V}_2$  gives another decomposition, whose sums of squares and degrees of freedom are sums of the old ones. The regression row is a coarsening of the table that gives each regressor its own row (Section 9.3). **Refining** splits a row: any subspace of dimension  $r_i$  can be cut into  $r_i$  orthogonal lines, each carrying one degree of freedom. A refinement is useful only when its lines answer questions of interest. Section 9.4 chooses them with orthogonal contrasts and orthogonal polynomials.

*Idea.* An analysis of variance table is a list of mutually orthogonal subspaces that fill  $\mathbb{R}^n$ . Its sums of squares are the squared lengths of the projections of  $\mathbf{y}$ , its degrees of freedom are the dimensions, and both columns add up because of Pythagoras.

## 9.1.6 Exercises

### 9.1.6.1 A. Check your understanding

#### Exercise A1.

For simple regression,  $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$  with  $\mathbf{x}$  not constant, show that  $\text{SSR} = \hat{\beta}_1^2 S_{xx} = S_{xy}^2 / S_{xx}$  and that the  $F$  ratio of the corrected table equals  $t^2$ , where  $t = \hat{\beta}_1 \sqrt{S_{xx}} / s$ .

#### Exercise A2.

A regression with an intercept has a corrected table in which the regression row has 3 degrees of freedom and sum of squares 120, the residual mean square is 2.5, and the corrected total sum of squares is 172.5. Find  $n$ , the residual degrees of freedom, the  $F$  ratio and  $R^2$ .

### 9.1.6.2 B. Practice

**Exercise B1.**

Let  $\mathbf{U} = [\mathbf{Q}_1, \dots, \mathbf{Q}_{k+1}]$  be the orthogonal matrix of [Theorem 9.1.1\(d\)](#) and  $\mathbf{z} = \mathbf{U}^\top \mathbf{Y}$ . Suppose  $E(\mathbf{Y}) = \boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ . Show that the coordinates of  $\mathbf{z}$  are uncorrelated with variance  $\sigma^2$ , that the last  $n - r$  of them have mean zero, and deduce  $E(\text{SSE}) = (n - r)\sigma^2$ . Show that if in addition  $\mathbf{Y}$  is normal, the coordinates are independent.

*Solution.* By Theorem 2.2.1,  $\text{Cov}(\mathbf{z}) = \mathbf{U}^\top (\sigma^2 \mathbf{I}) \mathbf{U} = \sigma^2 \mathbf{I}$  and  $E(\mathbf{z}) = \mathbf{U}^\top \boldsymbol{\mu}$ . The last block is  $\mathbf{Q}_{k+1}^\top \boldsymbol{\mu}$ , and  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$  is orthogonal to the error space, so it is zero. Then  $\text{SSE} = \|\mathbf{Q}_{k+1}^\top \mathbf{Y}\|^2$  is a sum of  $n - r$  squares of variables with mean zero and variance  $\sigma^2$ , so its expectation is  $(n - r)\sigma^2$ . Under normality  $\mathbf{z}$  is normal (Theorem 3.3.1) with diagonal covariance, and uncorrelated jointly normal variables are independent (Theorem 3.4.1).

**Exercise B2.**

For a model without an intercept, the quantity  $1 - \text{SSE}/\text{SST}$  is sometimes reported as  $R^2$ . Show that it is negative iff the model fits worse than the intercept-only model. Construct a data set with  $n = 4$  and one regressor for which it is below  $-1$ .

**9.1.6.3 C. Going deeper****Exercise C1.**

Let  $\mathcal{V}$  be a subspace with projection  $\mathbf{P}$ . Show that

$$\|\mathbf{P}\mathbf{y}\|^2 = \max_{\mathbf{v} \in \mathcal{V}, \mathbf{v} \neq \mathbf{0}} \frac{(\mathbf{v}^\top \mathbf{y})^2}{\mathbf{v}^\top \mathbf{v}},$$

with the maximum attained at  $\mathbf{v} = \mathbf{P}\mathbf{y}$  when  $\mathbf{P}\mathbf{y} \neq \mathbf{0}$ . Interpret: the sum of squares of a row with  $r_i$  degrees of freedom is the largest one-degree-of-freedom sum of squares obtainable by refining that row. [Chapter 13](#) uses this fact to build Scheffé's simultaneous intervals.

*Solution.* For  $\mathbf{v} \in \mathcal{V}$ ,  $\mathbf{v}^\top \mathbf{y} = \mathbf{v}^\top \mathbf{P}\mathbf{y}$ , so by Cauchy–Schwarz (Proposition 1.1.1)  $(\mathbf{v}^\top \mathbf{y})^2 \leq \mathbf{v}^\top \mathbf{v} \|\mathbf{P}\mathbf{y}\|^2$ , with equality iff  $\mathbf{v}$  is a multiple of  $\mathbf{P}\mathbf{y}$ . Taking  $\mathbf{v} = \mathbf{P}\mathbf{y} \in \mathcal{V}$  attains the bound. A line  $\text{span}(\mathbf{v})$  inside  $\mathcal{V}$  carries the one-degree-of-freedom sum of squares  $(\mathbf{v}^\top \mathbf{y})^2 / \mathbf{v}^\top \mathbf{v}$  (Example (Projections onto a line)), so no refinement of the row gives a line with a larger sum of squares than the whole row.